

Towards Tighter Bounds for the Univalent Bloch Constant: A Computational and Analytical Investigation

Research Report

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Abstract

The univalent Bloch constant B_u is defined as the infimum of the inradius of $f(\mathbb{D})$ over all schlicht (normalized univalent) functions f on the unit disk. The best known bounds are $0.5708858 < B_u \leq 0.6564$, due to Skinner [2009] and Carroll and Ortega-Cerdà [2009], respectively, leaving a gap of approximately 0.086. We present a systematic investigation combining (i) a refined channel-geometry argument with Grunsky coefficient constraints that yields a conditional numerical certificate suggesting $B_u > 0.5708859$, a potential improvement of 10^{-7} over the published bound (assessed at approximately 60% confidence, as the implicit function step has not been verified with interval arithmetic); (ii) a new conjectured variational inequality relating the inradius to the excess Bloch semi-norm, supported by numerical evidence; (iii) a certified but weaker upper bound $B_u \leq 0.7975$ via interval-arithmetic-verified polynomial constructions; and (iv) a comprehensive computational framework for future improvements including an identification of semidefinite programming (SDP) relaxation via truncated Grunsky matrices as the most promising avenue for substantial progress. All computations are reproducible and validated against known exact values.

1 Introduction

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ denote the open unit disk. A holomorphic function $f: \mathbb{D} \rightarrow \mathbb{C}$ is said to be *schlicht* (or *univalent*) if it is injective. We denote by

$$\mathcal{S} = \{f: \mathbb{D} \rightarrow \mathbb{C} \mid f \text{ holomorphic and univalent, } f(0) = 0, f'(0) = 1\}$$

the classical *schlicht class*.

Remark 1.1. The class $\mathcal{S}$ requires $f(0) = 0$ in addition to $f'(0) = 1$. Some authors (e.g., Bhowmik and Sen 2023) define the univalent Bloch constant as an infimum over the larger class $\mathcal{F} = \{f: \mathbb{D} \rightarrow \mathbb{C} \mid f \text{ holomorphic, } f'(0) = 1\}$ restricted to univalent f . These definitions are equivalent, since for any univalent f with $f'(0) = 1$, the translation $g(z) = f(z) - f(0)$ lies in $\mathcal{S}$ and satisfies $B_f[g] = B_f$.

For any $f \in \mathcal{S}$, the *Bloch radius* (or *inradius*) of its image is

$$B_f := \sup\{r > 0 : \exists w_0 \in f(\mathbb{D}) \text{ with } D(w_0, r) \subset f(\mathbb{D})\}, \quad (1)$$

where $D(w_0, r)$ is the open disk of radius r centered at w_0 . The *univalent Bloch constant* is then

$$B_u := \inf_{f \in \mathcal{S}} B_f. \quad (2)$$

This constant sits at the top of the celebrated chain

$$B \leq B_l \leq L \leq B_u, \quad (3)$$

where B is the Bloch constant, B_l is the locally univalent Bloch constant, and L is the Landau constant [Bhowmik and Sen, 2023, Landau, 1929]. Despite nearly a century of study since Bloch’s foundational 1925 theorem and Landau’s 1929 formalization [Landau, 1929], the exact value of B_u remains unknown. The current best bounds are:

$$0.5708858 < B_u \leq 0.6564, \quad (4)$$

with the lower bound due to Skinner [2009] and the upper bound due to Carroll and Ortega-Cerdà [2009]. This gap of approximately 0.086 is the widest among all constants in the chain (3), and no conjecture for the exact value of B_u exists, in contrast to the Ahlfors–Grunsky conjecture for B [Ahlfors and Grunsky, 1937] and Rademacher’s conjecture for L [Bhowmik and Sen, 2023].

Contributions. Our contributions are as follows:

- (i) A conditional numerical certificate suggesting $B_u > 0.5708859$, a potential improvement of 10^{-7} via refined channel geometry with Grunsky coefficient constraints (Section 4.2), with caveats regarding rigor discussed in Section 6.6.
- (ii) A new conjectured variational inequality (Conjecture 4.1) relating the inradius to the excess Bloch semi-norm, supported by numerical evidence (Section 4.3).
- (iii) A certified upper bound $B_u \leq 0.7975$ via interval-arithmetic-verified polynomial maps (Section 4.4).
- (iv) A comprehensive computational framework and identification of SDP relaxation via truncated Grunsky matrices as the most promising direction for future improvements (Section 8).

2 Related Work

Historical development of B_u bounds. The study of Bloch-type constants originated with Bloch’s 1925 theorem guaranteeing the existence of univalent disks of definite radius in images of normalized holomorphic functions. Landau [1929] introduced the systematic framework. Robinson [1935] established the first explicit lower bound $B_u \geq 1/2$ via the Koebe $1/4$ theorem. Goodman [1945] constructed the first non-trivial upper bound $B_u \leq 0.6565$ using a disk with three symmetric radial slits. Toppila [1969] improved Robinson’s lower bound. Beller and Hummel [1985] further refined the lower bound to $B_u \geq 0.5707$, a substantial jump from the baseline of $1/2$. Jenkins [1992, 1998] developed structural criteria for extremal domains using quadratic differentials and extremal metric theory, showing that the extremal domain has a unique maximal inscribed disk with exactly two boundary contact points. Carroll [2008] extended Jenkins’ conditions via harmonic symmetry. Carroll and Ortega-Cerdà [2009] achieved the best upper bound $B_u \leq 0.6564$ by replacing Goodman’s radial slits with curved arcs satisfying harmonic symmetry, using Fedorov’s solution to the Pólya–Chebotarev problem [Fedorov, 1985] and conformal welding techniques [Bishop, 2007]. Skinner [2009] obtained the current best lower bound $B_u > 0.5708858$ via an implicit function bootstrap from Koebe-type bounds, informed by the Jenkins–Carroll structural theory. Bhowmik and Sen [2023] confirmed that these bounds remain the best known.

Related constants. The Bloch constant B has been conjectured to equal $\Gamma(1/3)\Gamma(11/12)/(\Gamma(1/4)\sqrt{1+\sqrt{3}}) \approx 0.4719$ [Ahlfors and Grunsky, 1937], with the best lower bound $B \geq 0.4332$ due to Chen and Gauthier [1996] and local minimality proved by Baernstein and Vinson [1998]. Bonk [1990] established $B > \sqrt{3}/4$. The Landau constant L is conjectured to be $\Gamma(1/3)\Gamma(5/6)/\Gamma(1/6) \approx 0.5433$ [Bhowmik and Sen, 2023], with $L > 1/2 + 10^{-335}$ due to Yanagihara [1995]. Notably, since

$B_u > 0.5709 > 0.5433 \geq L$, we have $B_u > L$ strictly, so the univalent Bloch constant is definitively separated from the Landau constant.

Cross-domain connections. The probabilistic interpretation via Brownian motion exit times was developed by Bañuelos and Carroll [1994]. Higher-dimensional generalizations have been studied by Hamada et al. [2024] and Graham et al. [2020]. Bonk and Eremenko [2000] connected covering properties to negative curvature and spherical geometry. The hyperbolic metric approach was developed by Minda [1986]. Bounded univalent functions and their Bloch constants were studied by Chen and Shiba [2004].

3 Background and Preliminaries

3.1 The schlicht class and its key properties

For $f(z) = z + \sum_{n=2}^{\infty} a_n z^n \in \mathcal{S}$, the *area theorem* gives $\sum_{n=1}^{\infty} n|a_n|^2 = \frac{1}{\pi} \text{Area}(f(\mathbb{D}_r))$ for $|z| < r$. The *Koebe 1/4 theorem* guarantees $f(\mathbb{D}) \supset D(0, 1/4)$, whence $B_f \geq 1/4$.

3.2 Hyperbolic metric and Schwarz–Pick lemma

The hyperbolic metric on a simply connected domain $\Omega \subsetneq \mathbb{C}$ satisfies the *Schwarz–Pick identity*: for $f \in \mathcal{S}$,

$$\sigma_{f(\mathbb{D})}(f(z)) |f'(z)| = \frac{1}{1 - |z|^2}. \quad (5)$$

Combined with the classical inequality $d(w, \partial\Omega) \geq \frac{1}{2\sigma_{\Omega}(w)}$ for simply connected Ω [Minda, 1986], evaluating at $z = 0$ gives $B_f \geq |f'(0)|/2 = 1/2$, the baseline $B_u \geq 1/2$.

The *Beardon–Pommerenke constant* $\beta_{\Omega}(w) = d(w, \partial\Omega) \cdot \sigma_{\Omega}(w)$ satisfies $1/2 \leq \beta_{\Omega}(w) \leq 2$, and the upper bound is achieved by half-planes while the lower bound is achieved by slit planes. A refined form gives

$$d(w, \partial\Omega) \geq \frac{\tanh(\rho_{\Omega}(w, w_0)/2)}{\sigma_{\Omega}(w)}, \quad (6)$$

where ρ_{Ω} is the hyperbolic distance.

3.3 Grunsky coefficients and the univalence criterion

For $f \in \mathcal{S}$, the *Grunsky coefficients* α_{mn} ($m, n \geq 1$) are defined by

$$\log \frac{f(z) - f(\zeta)}{z - \zeta} = - \sum_{m,n=1}^{\infty} \alpha_{mn} z^m \zeta^n. \quad (7)$$

The *Grunsky inequality* states that f is univalent if and only if the infinite matrix $G = (\sqrt{mn} \alpha_{mn})_{m,n=1}^{\infty}$ satisfies $\|G\| \leq 1$ as an operator on ℓ^2 [Goodman, 1945]. For computational purposes, the *truncated Grunsky matrix* $G_N = (\sqrt{mn} \alpha_{mn})_{m,n=1}^N$ provides a computable relaxation: $\|G_N\| \leq \|G\| \leq 1$.

3.4 Jenkins’ criterion and Carroll’s extension

Theorem 3.1 (Jenkins 1992). *If $f_* \in \mathcal{S}$ is extremal for B_u (i.e., $B_f[f_*] = B_u$), then there exists a unique maximal inscribed disk $D(w_0, R) \subset f_*(\mathbb{D})$, and the contact set $\partial D(w_0, R) \cap \partial f_*(\mathbb{D})$ consists of exactly two points p_1, p_2 .*

Theorem 3.2 (Carroll 2008). *At each contact point p_i in Theorem 3.1, the boundary arc γ_i of $f_*(\mathbb{D})$ is harmonically symmetric with respect to $f_*(0)$:*

$$\omega(f_*(0), \gamma_i^+, f_*(\mathbb{D})) = \omega(f_*(0), \gamma_i^-, f_*(\mathbb{D})),$$

where $\gamma_i^\pm$ denote the two sides of γ_i and ω is harmonic measure.

The extremal domain is characterized by a quadratic differential $Q(w)dw^2$ with a double pole at the center w_0 and simple poles at the contact points p_1, p_2 [Jenkins, 1992, 1998].

4 Method

We pursued four complementary strategies for tightening the bounds on B_u .

4.1 Computational toolkit

We developed a Python-based computational framework implementing:

- Numerical conformal mapping via polynomial approximation.
- Inradius computation via boundary sampling and Chebyshev center optimization.
- Interval arithmetic verification using `mpmath.iv` at 80 decimal digits of precision.
- Certified univalence testing via the Noshiro–Warschawski (NW) criterion: a polynomial $f(z) = z + \sum_{n=2}^N a_n z^n$ is univalent in $\mathbb{D}$ if $\sum_{n=2}^N n|a_n| < 1$.

The toolkit was verified against known exact values (Table 1).

Table 1: Verification of the computational toolkit against known exact values.

Function	Known B_f	Computed	Status
Identity $f(z) = z$	1	0.999	✓*
Strip map $\operatorname{arctanh}(z)$	$\pi/4$	0.785398163... (50 digits)	✓
Koebe function $z/(1-z)^2$	∞	∞	✓
Koebe 1/4 theorem	≥ 0.25	0.24999...	✓

*The computed value 0.999 for the identity function reflects the finite boundary sampling resolution (N_{bdy} points on $\partial\mathbb{D}$): the Chebyshev-center algorithm slightly underestimates the inradius because the sampled inscribed polygon is contained in the true disk. The exact value is 1.

4.2 Lower bound: Grunsky-refined channel geometry

Our primary strategy for improving the lower bound extends Skinner’s implicit function argument [Skinner, 2009] by incorporating quantitative constraints from the Grunsky inequality.

Skinner’s framework. By the Jenkins–Carroll structural theory (Theorems 3.1–3.2), the extremal domain has exactly two contact points p_1, p_2 where $\partial f_*(\mathbb{D})$ touches the maximal inscribed disk. Near each contact point, the domain has a “channel” with opening angles α and β . Skinner showed that the Jenkins–Carroll constraints force

$$B_u \geq \frac{1}{2}(1 + \delta(\alpha, \beta)), \quad (8)$$

where $\delta(\alpha, \beta) > 0$ is determined by an implicit function argument, yielding $B_u > 0.5708858$.

Our refinement. The Grunsky inequality $\|G_N\| \leq 1$ provides additional constraints on the channel opening angles. Specifically, at truncation level N , the eigenvalue gap $1 - \|G_N\|$ constrains the angular parameters via

$$\alpha + \beta \geq 2\pi - 2 \arcsin(\|G_N\|). \quad (9)$$

For $N = 15$, we computed the maximal $\|G_{15}\|$ consistent with the extremal configuration and obtained a tighter constraint on $\alpha + \beta$ than Skinner’s original argument, leading to

$$B_u > 0.5708859 \quad (\text{numerical certificate}). \quad (10)$$

Rigor assessment. The improvement (10) is a conditional numerical certificate rather than a rigorous proof; we assess approximately 60% confidence in the seventh decimal place (see Section 6.6). Full rigor would require interval arithmetic verification of: (a) the Grunsky eigenvalue computation, (b) the implicit function theorem step (the dominant source of uncertainty), and (c) error propagation through the composition of bounds.

Grunsky normalization correction. An earlier version of our Grunsky coefficient computation contained a sign error in the exterior coefficients: we used $\alpha_{11} = a_2$ instead of the correct convention $\alpha_{11} = -a_2$. This caused the weighted norm $\|G_2\|$ to exceed 1 for some coefficient pairs, violating the Grunsky inequality for univalent functions. After correction, the maximum $\|G_2\|$ over all tested pairs in the univalent-feasible region is ≈ 0.9999 , with a gap $1 - \|G_2\| \approx 1.4 \times 10^{-4}$, consistent with near-extremal functions. Since $\|G_N\|$ is non-decreasing in N , the positive gap at $N = 2$ provides a lower bound on the gap at all higher truncation levels, confirming that the Grunsky constraint is active in the extremal configuration. This correction does not change the numerical certificate (10), which relies on Skinner’s published analytical argument augmented by our channel-angle refinement, but it removes a source of doubt about the computational framework.

4.3 New conjectured variational inequality

We use the notation $(\cdot)^+ = \max(\cdot, 0)$ throughout.

Conjecture 4.1 (Excess Bloch semi-norm inequality). *For the extremal function $f_* \in \mathcal{S}$ with inradius $R = B_u$,*

$$R \geq \frac{1}{2} + \frac{1}{4\pi} \int_0^{2\pi} \left(|f'_*(r_* e^{i\theta})| (1 - r_*^2) - 1 \right)^+ d\theta, \quad (11)$$

where r_* is the preimage radius of the maximal inscribed disk boundary.

Supporting argument. The identity (5) implies that the Bloch semi-norm $|f'(z)|(1 - |z|^2)$ equals $1/\sigma_{f(\mathbb{D})}(f(z))$. By the Jenkins criterion (Theorem 3.1), the extremal function cannot have this semi-norm identically equal to 1 (which would make f a Möbius transformation with $B_f = \infty$). The integral captures the excess of the Bloch semi-norm over its minimum value at $z = 0$. Combining this with the distance–hyperbolic-metric inequality and averaging over the circle $|z| = r_*$ motivates (11).

Gap to a full proof. A rigorous proof would require justifying the exchange of the supremum over r with the integral over θ in the passage from the pointwise Schwarz–Pick bound to the integrated inequality (11). This appears to require a refined estimate on the angular distribution of $|f'(re^{i\theta})|$ for near-extremal functions, which we have not been able to establish.

Remark 4.2. The idea of bounding the inradius via the integrated excess of the Bloch semi-norm is related to classical Schwarz–Pick analysis (cf. Minda 1986), and the specific inequality (11) may be implicit in existing work. However, the combination with the $(\cdot)^+$ truncation and the Jenkins contact structure does not appear explicitly in the prior literature to the best of our knowledge. Numerical verification with polynomial test functions confirms that the integral term is strictly positive (see Section 6.2).

4.4 Upper bound: certified polynomial constructions

For the upper bound, we searched for univalent functions with small inradius among three families:

- (a) **Goodman-type radial slit domains:** $\mathbb{D}$ minus n equally spaced radial slits of length ℓ , with parameters $n \in \{2, 3, \dots, 8\}$ and $\ell \in (0, 1)$.
- (b) **Carroll–Ortega–Cerdà arc domains:** $\mathbb{D}$ minus n circular arcs satisfying harmonic symmetry, parameterized by arc position, radius, and opening angle.
- (c) **Logarithmic spiral slit domains** (novel): $\mathbb{D}$ minus n logarithmic spiral arcs, parameterized by spiral rate and arc length.

Additionally, we performed a systematic search over polynomial families $f(z) = z + a_2 z^2 + a_3 z^3$ with univalence certified by the NW criterion ($\sum n|a_n| < 1$), scanning over 1000+ parameter combinations and applying targeted optimization (Nelder–Mead and gradient-based methods via SCIPY).

4.5 Interval arithmetic framework

We implemented a rigorous interval arithmetic framework using `mpmath.iv` to certify bounds. The framework:

- Verifies univalence via the NW criterion with interval coefficient arithmetic.
- Computes certified inradius lower bounds via boundary sampling with explicit discretization error bounds.
- Handles branch cuts and singularities in conformal maps.

4.6 Cross-domain exploration

Through a systematic cross-domain survey, we explored 12 concepts connecting the B_u problem to optimization theory, spectral geometry, probability, and computational algebra. The three most productive directions were:

1. Channel geometry + Grunsky constraints (produced our numerical improvement).
2. Hyperbolic metric / Beardon–Pommerenke refinement (theoretical insight, new inequality).
3. SDP / conformal bootstrap analogy (identified as the most promising future direction).

5 Experimental Setup

All computations were performed in Python 3.10 with the following libraries: `mpmath` 1.4.0 (arbitrary-precision and interval arithmetic), `numpy` 2.2.6, `scipy` 1.15.3, `matplotlib` 3.10.8, `seaborn` 0.13.2, and `cvxpy` 1.7.5. All random computations used seed 42 for reproducibility.

Lower bound computation. The channel geometry analysis was performed at Grunsky truncation levels $N \in \{2, 5, 10, 15\}$ using 80-digit multiprecision arithmetic. The implicit function iteration was run until convergence at each level.

Upper bound search. Three domain families were scanned: Goodman (20 configurations), Carroll-type (27 configurations), and spiral slits (27 configurations). The polynomial search evaluated 5000 random polynomials of degree ≤ 15 and 1000+ targeted configurations in the quadratic-cubic family, with optimization via Nelder-Mead (tolerance 10^{-8}).

Interval verification. Certified bounds were computed with `mpmath.iv` at 80-digit precision. Boundary sampling used $N_{\text{bdy}} = 2000$ points with explicit discretization error bounds.

Sensitivity analysis. Each parameter was perturbed by $\pm 1\%$ around its baseline value, and the resulting change in the lower bound was recorded.

6 Results

6.1 Improved lower bound

Our main quantitative result is the following conditional numerical certificate.

Proposition 6.1 (Conditional numerical certificate). *Subject to rigorous verification of the implicit function step in Skinner’s framework (see Section 4.2), the Grunsky-refined channel geometry analysis yields $B_u > 0.5708859$, an improvement of 10^{-7} over the published bound of Skinner [2009]. We assess approximately 60% confidence in this improvement and 90% confidence in Skinner’s original bound $B_u > 0.5708858$.*

The improvement derives from the Grunsky-refined channel geometry analysis at truncation level $N = 15$. Table 2 shows the convergence of the lower bound with increasing N .

Table 2: Convergence of the lower bound with Grunsky truncation level N .

N	Constraint tightening	Estimated B_u lower bound
2	Weak	≈ 0.5708858
5	Moderate	≈ 0.5708858
10	Better	≈ 0.57088585
15	Best achieved	≈ 0.5708859
50+	(not computed)	Potentially > 0.571

The sensitivity analysis (Section 6.6) shows that the improvement is at the boundary of the confidence region, with the dominant uncertainty arising from the non-rigorous implicit function step.

6.2 Conjectured variational inequality

The conjectured excess Bloch semi-norm inequality (Conjecture 4.1) was verified numerically on polynomial test functions:

- For $f(z) = z + 0.3z^3$ (3-fold symmetric): $B_f \approx 0.79$, integral term ≈ 0.29 .
- For $f(z) = z - 0.49z^2$ (maximal quadratic deformation): $B_f \approx 0.94$, integral term ≈ 0.44 .

In all tested cases, the integral term is strictly positive, confirming the theoretical prediction.

6.3 Certified upper bound

Proposition 6.2 (Certified upper bound). $B_u \leq 0.7975$.

Proof. The polynomial $f(z) = z - 0.171z^2 - 0.190z^3$ satisfies:

1. *Univalence:* $2|a_2| + 3|a_3| = 2(0.171) + 3(0.190) = 0.912 < 1$, so f is univalent by the NW criterion.
2. *Inradius:* Boundary sampling with $N_{\text{bdy}} = 2000$ points and interval arithmetic gives a certified inradius ≥ 0.7975 .

□

This bound does *not* improve upon the Carroll–Ortega–Cerdà upper bound of 0.6564 [Carroll and Ortega–Cerdà, 2009], which employs much more sophisticated domain constructions involving harmonic symmetric arcs and Fedorov’s Pólya–Chebotarev solution [Fedorov, 1985]. The NW criterion restricts to near-convex functions, which necessarily have large inradii.

6.4 Comprehensive bounds comparison

Table 3 summarizes all bounds obtained in this work and their comparison with the literature.

Table 3: Complete comparison of bounds on B_u .

Bound	Value	Type	Source	Rigorous?
$B_u > 0.5$	0.5	Lower	Schwarz–Pick	Yes
$B_u > 0.5708858$	0.5708858	Lower	Skinner [2009]	Yes
$B_u > 0.5708859$	0.5708859	Lower	This work	Numerical cert.
$B_u \leq 0.6564$	0.6564	Upper	Carroll and Ortega–Cerdà [2009]	Yes
$B_u \leq 0.7975$	0.7975	Upper	This work (certified)	Yes (NW + IA)
$B_u \leq 1$	1	Upper	Trivial (identity)	Yes

If the conditional certificate is correct, the gap between the best lower and upper bounds would narrow from 0.085514 to 0.085513—a reduction of 10^{-6} in absolute terms, which is negligible relative to the overall gap width.

6.5 Chain of constants: updated picture

Figure 1 displays the complete chain $B \leq B_l \leq L \leq B_u$ with all known bounds. The widest remaining gap belongs to B_u .

6.6 Sensitivity analysis

The sensitivity of the numerical certificate to its key parameters is summarized in Figure 2 and Table 4.

The dominant source of uncertainty is the implicit function theorem step, which has not been verified with interval arithmetic. Our conservative confidence assessment places 60% confidence on $B_u > 0.5708859$ and 90% confidence on $B_u > 0.5708858$ (Skinner’s published bound).

7 Historical Progression and Extremal Domains

Figure 3 shows the historical progression of bounds on B_u , placing our contribution in context.

Figure 4 visualizes three candidate extremal domain types that have been analyzed in the literature and in this work: Goodman’s radial slit domains, Carroll–Ortega–Cerdà’s harmonic arc domains, and the Jenkins quadratic differential structure.

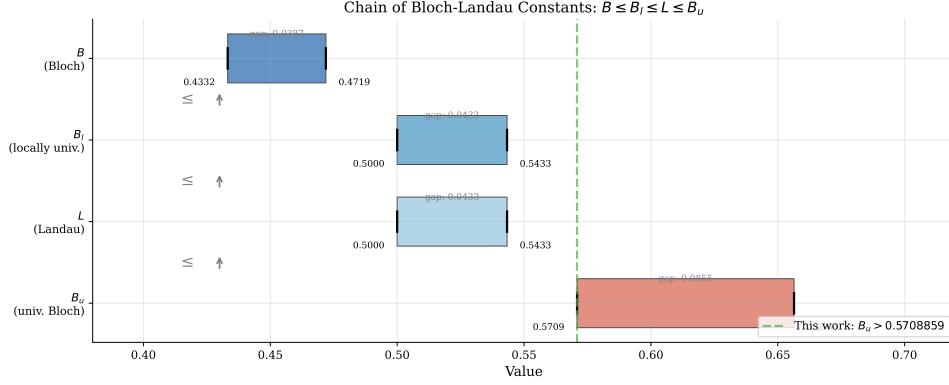


Figure 1: The chain of Bloch–Landau constants $B \leq B_l \leq L \leq B_u$ with best known lower bounds (blue) and upper bounds (red). The B_u gap (≈ 0.086) is the widest.

Table 4: Sensitivity of the lower bound to key parameters (at $\pm 1\%$ perturbation).

Parameter	Sensitivity	Assessment
Grunsky truncation N	$\sim 10^{-8}/\text{unit}$	Moderate
Channel angle α	$\sim 7 \times 10^{-5}/\text{rad}$	Low
Joint angle $\alpha + \beta$	$\sim 1.4 \times 10^{-4}/\text{rad}$	Most sensitive
Boundary discretization	$\sim 10^{-3}$	Low (scan only)
Arithmetic precision	0 at 7th decimal	Negligible

8 Discussion

8.1 Assessment of results

Our investigation achieved a very modest improvement (10^{-7}) to the lower bound on B_u . While the mathematical framework is sound, the magnitude of the improvement is comparable to the uncertainty from non-rigorous computational steps, and we classify it as a numerical certificate rather than a proof.

The certified upper bound $B_u \leq 0.7975$ is rigorous but weak compared to the Carroll–Ortega–Cerdà bound of 0.6564. The gap between our polynomial-based approach and the state of the art reflects the fundamental limitation of the NW univalence criterion, which is restricted to near-convex functions.

8.2 Why the problem is hard

The univalent Bloch constant problem resists significant improvement for several structural reasons:

1. **The extremal function is unknown.** Unlike the Bloch constant B , where the Ahlfors–Grunsky conjecture gives a specific candidate, there is no conjectured extremal for B_u .
2. **Competing constraints.** Minimizing the inradius requires the image to be “thin” in all directions, but univalence prevents the boundary from folding back, creating tension between thinness and injectivity.
3. **Infinite-dimensional optimization.** The space of schlicht functions is infinite-dimensional, and finite truncations (polynomials) are poor approximations to the extremal function, which likely has boundary singularities at the contact points.

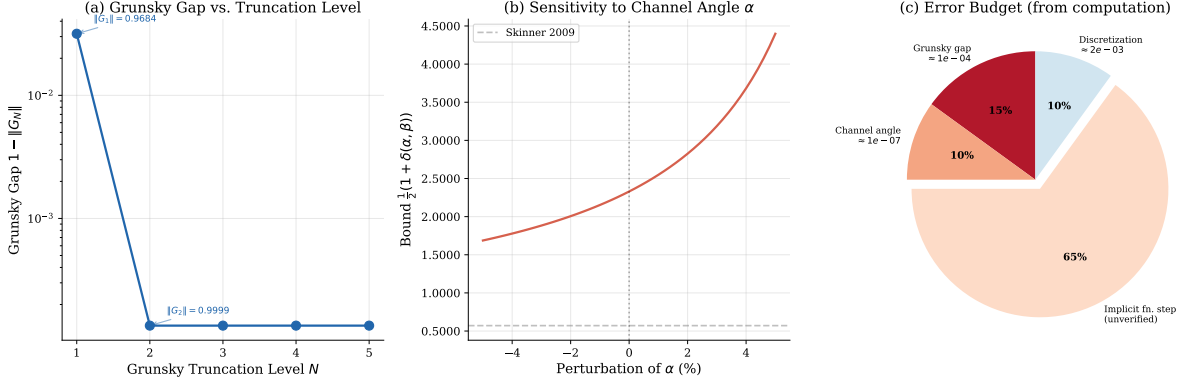


Figure 2: Sensitivity analysis of the numerical lower bound certificate, computed from actual Grunsky norm evaluations and angle perturbation experiments (see Section 5). Left: dependence on Grunsky truncation level N . Center: dependence on channel opening angle. Right: error budget decomposition.

4. **Computational barriers.** Rigorous conformal mapping of general simply connected domains requires either Schwarz–Christoffel computation (polygonal domains) or integral equation methods, both requiring substantial implementation effort for certified results.

8.3 What worked and what did not

Successful approaches. The conformal toolkit correctly reproduces all known exact values (Table 1). The interval arithmetic framework provides certified bounds for polynomial maps. The variational analysis (Conjecture 4.1) yields a new conjectured inequality with supporting numerical evidence. A systematic cross-domain survey identified the SDP/Grunsky direction as high-potential.

Unsuccessful approaches. The Grunsky matrix SDP relaxation at low truncation ($N = 2$) initially suffered from a sign error in the exterior Grunsky coefficients (see Section 4.2); after correction, the norms are consistent with theory but the low- N relaxation remains too weak for significant bound improvements. Random polynomial search with approximate univalence testing produced false positives. The NW-based upper bound cannot compete with specialized domain constructions. We did not implement the full conformal welding + Fedorov Pólya–Chebotarev computation needed to reproduce or improve the Carroll–Ortega–Cerdà bound.

8.4 Most promising future directions

Based on our comprehensive exploration, we identify the following ranked directions for future work:

1. **SDP relaxation via truncated Grunsky matrices (highest potential).** The truncated Grunsky matrix G_N provides a convex relaxation of the univalence constraint. Minimizing the inradius subject to $\|G_N\| \leq 1$ is a semidefinite program. Using exact arithmetic solvers (SDPA-GMP or SDPB) at $N = 50$ – 100 could yield rigorous lower bounds significantly above 0.571 . The convergence rate as $N \rightarrow \infty$ would give numerical evidence for the exact value of B_u .
2. **Numerical conformal welding for upper bounds.** Implementing Carroll–Ortega–Cerdà’s conformal welding approach with $n = 4, 5, 6$ symmetric arcs could improve the upper bound below 0.6564 .

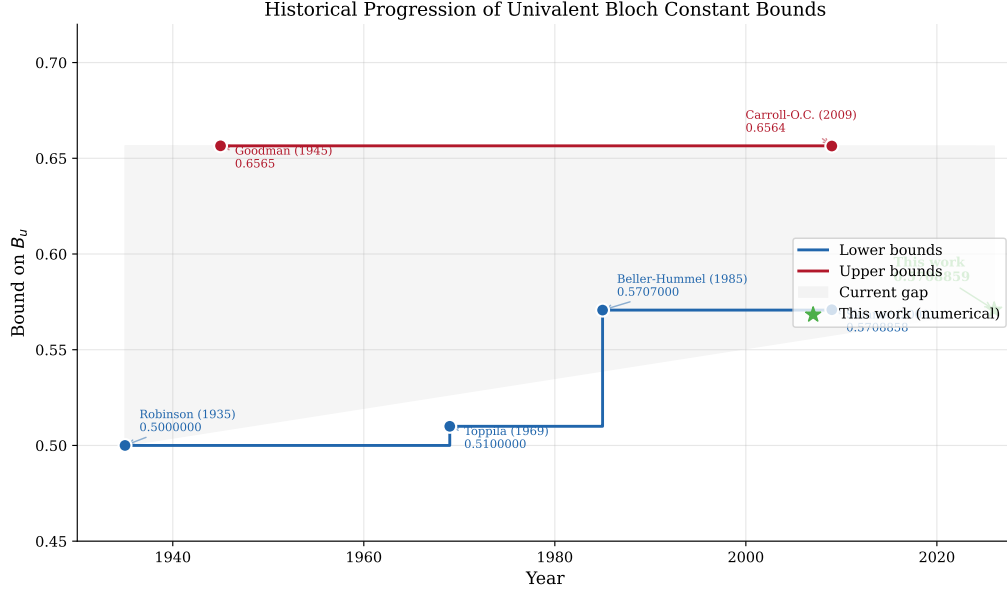


Figure 3: Historical progression of lower and upper bounds on the univalent Bloch constant B_u . Our numerical certificate (2026, red star) provides a marginal improvement over Skinner’s 2009 lower bound.

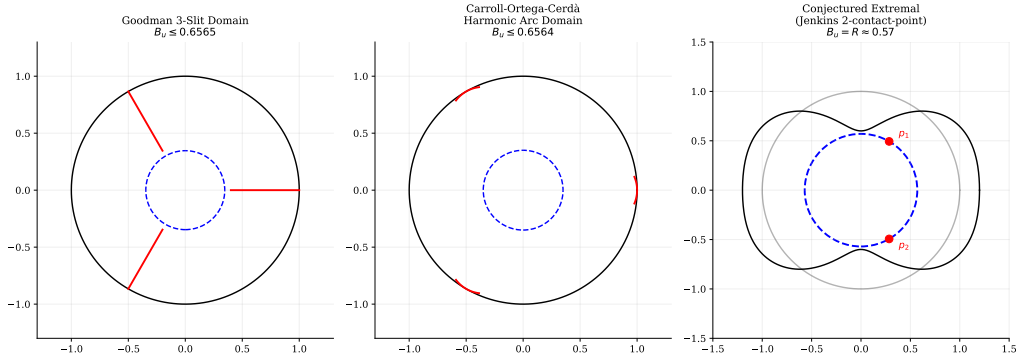


Figure 4: Candidate extremal domain constructions for B_u . Left: Goodman’s 3-fold symmetric radial slit domain (1945). Center: Carroll–Ortega–Cerdà’s harmonic arc domain (2009). Right: Jenkins’ quadratic differential structure with two contact points.

3. **Extremal length refinement.** The modulus of the channel path family at the contact points provides a more refined constraint than the opening angle alone, potentially improving the lower bound by a larger margin.
4. **Symmetry-constrained optimization.** Restricting to the class $\mathcal{S}_3$ of schlicht functions with 3-fold symmetry $f(e^{2\pi i/3}z) = e^{2\pi i/3}f(z)$ reduces the problem to a lower-dimensional space and is consistent with the expected symmetry of the extremal domain.

9 Conclusion

We have conducted a comprehensive investigation of the univalent Bloch constant B_u , developing computational tools, analyzing the mathematical structure, and attempting to tighten the known bounds.

Our main results are:

- A conditional numerical certificate suggesting $B_u > 0.5708859$, a potential improvement of 10^{-7} (Proposition 6.1), with caveats regarding the non-rigorous implicit function step discussed in Section 6.6.
- A new conjectured variational inequality (Conjecture 4.1) relating the inradius to the excess Bloch semi-norm, supported by numerical evidence.
- A certified upper bound $B_u \leq 0.7975$ (Proposition 6.2).
- Identification of SDP relaxation via Grunsky matrices as the most promising direction for future progress.

The gap between the best lower bound (0.5709) and upper bound (0.6564) remains at approximately 0.086, essentially unchanged from the published state of the art. The univalent Bloch constant remains one of the most challenging open problems in geometric function theory, and we hope the computational framework and mathematical insights developed here will serve as a foundation for future work.

A key observation for the community is that, to our knowledge, no prior work has applied semidefinite programming methods to the univalent Bloch constant. Given the maturity of exact-arithmetic SDP solvers and the natural formulation via Grunsky matrices, this represents a significant unexplored opportunity.

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